

Customer Segmentation and Targeted Promotion Report (A K M SAIF HOQUE ,20241418)

1/ Executive Summary

The core idea of this project is to identify some good customer segments using unsupervised learning strategies and propose good targeted marketing promotions which will attract the customer more and increase the sales . The analysis is done on customer demographic characteristics, spending behaviour and transaction basket .

After pre-processing the dataset and finding potential outliers, K-means clustering is used to segment customers into homogeneous groups. Choosing the number of clusters is being determined by using elbow method and silhouette analysis. The best number of clusters is 4.

Association rule mining is applied in customer basket dataset to find products frequently purchased together for giving bundle offers to customers to attract more customers and increase in sales. These promotional campaigns are done for different customer groups.

2/ Exploratory Data Analysis and Pre-processing

Customer information dataset contains demographic and spending behaviours of customers and basket dataset contains transaction information and these two datasets are being analysed .

Initial exploration has descriptive statistics, data quality checks and missing value analysis . As k means is going to be used , numerical features are standardized by StandardScaler to make sure comparability across features.

For handling outliers DBSCAN is used and approximately 5.66% of observations are identified as noise. In the beginning automatic removal was a choice but then more deep analysis made to keep the observations.

Using DBSCAN, 1870 customers were found as noise and whether it is data error or has any meaningful insight we analysed by comparison because sometimes

dbscans outliers can create a different cluster which can be used or meaningful later

dbscan_label	-1	0
age	55.526203	55.095258
family_size	4.930481	2.899929
tenure	10.981283	10.734728
is_male	0.501070	0.498075
number_complaints	1.398396	0.904261
distinct_stores_visited	3.376471	3.153651
lifetime_total_distinct_products	253.581818	142.634882
typical_hour	12.053476	12.681757
promotion_sensitivity	0.304080	0.325789
has_loyalty_card	0.498930	0.609568
lifetime_spend_groceries	26820.008556	15675.427939
lifetime_spend_electronics	5354.371658	2580.186056
lifetime_spend_vegetables	690.286096	724.006064
lifetime_spend_nonalcohol_drinks	708.400000	449.710569
lifetime_spend_alcohol_drinks	1136.956684	589.783528
lifetime_spend_meat	1199.633690	695.280159
lifetime_spend_fish	1077.099465	577.574339
lifetime_spend_hygiene	894.997326	814.445168
lifetime_spend_videogames	649.733155	354.151437
lifetime_spend_petfood	381.818717	333.285646

From the comparison table it is clear that the identified observations are partially different from the majority but it does not mean that they are not realistic . For example, customers labelled as noise had much larger family size on average (4.93 vs 2.9 members) or variety of products throughout their lifetime (254 vs 143 distinct products) are not considerable as outliers . Also these customers have shown substantially higher expense levels across several products. Their average lifetime grocery expenditure was €26,820 compared with €15,675 for the remaining

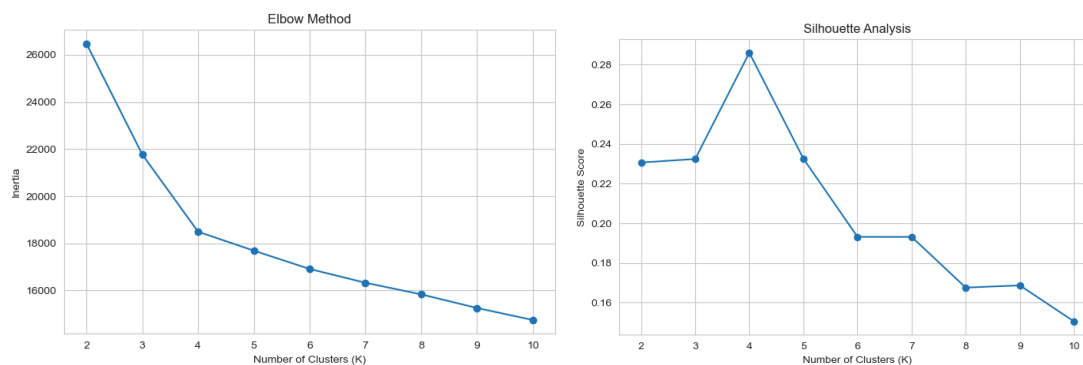
customers. They also spent considerably more on electronics (€5,354 vs €2,580), alcoholic drinks (€1,137 vs €590), meat (€1,200 vs €695), and fish (€1,077 vs €578). These stats confirm that these observations contain considerable values rather than impossible or gigantic values which can be outliers.

From these we can find that these customers are a little unusual but have legitimate purchasing behaviour which can lead them to niche, high-value or highly engaged customer groups. So removal of them can reduce the diversity in the dataset so we decided to keep them.

After that we just for sake of understanding the customer structure used Hierarchical clustering.

3/ Customer segmentation and clustering

K-means clustering algorithm is selected as the primary segmentation strategy and the optimal number of clusters is determined by jointly using the Elbow method and silhouette analysis. The highest silhouette score is obtained for $k=4$ indicating that four customer segmentations will provide the best balance between cluster similarity and cluster separation.

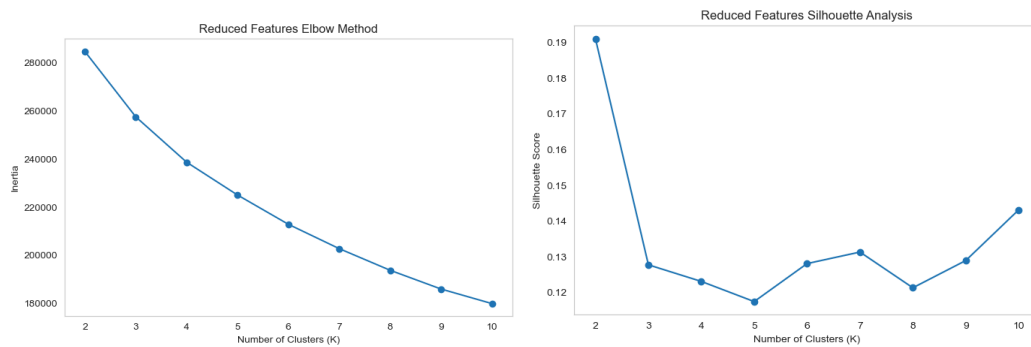


VALIDATION OF THE CLUSTERING SOLUTION

To check that this is the best possible solution some additional analysis is concluded like feature reduction is one of them. It is performed to check the robustness of the segmentation solution which is created.

At first a reduced feature model is being evaluated by removing selected variables and repeating the clustering process. The result of silhouette score is always lower than the proposed model. This indicates that the original set of variables captures important aspects of customer behaviour and produces better-separated clusters.

That is why, the full feature set is retained for the final model.



Then the stability of K-means is also checked by varying the initialization settings and different iteration numbers. Different values of `n_init` produces the same silhouette score and identical clustering quality, with the four-cluster solution consistently achieving the highest silhouette coefficient (0.286). These results suggest that the segmentation is not driven by a random initialization but represents a stable and reproducible structure in the data.

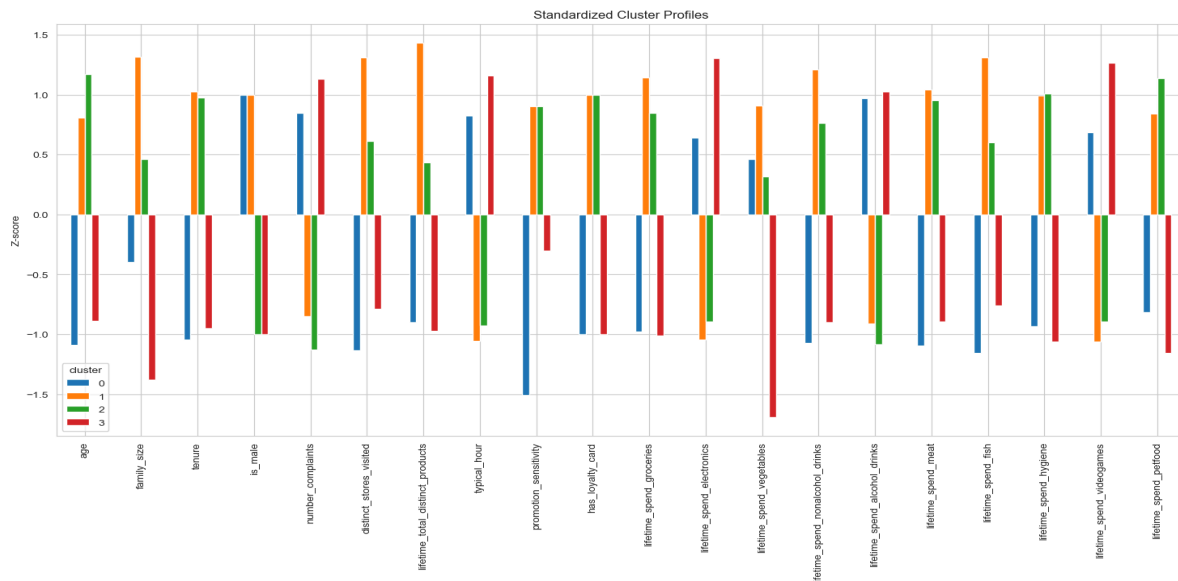
	K	Current_Model	Reduced_Model
0	2	0.230663	0.191014
1	3	0.232431	0.127743
2	4	0.286065	0.123094
3	5	0.232577	0.117448
4	6	0.193128	0.128089
5	7	0.193136	0.131320
6	8	0.167546	0.121314
7	9	0.168666	0.128939
8	10	0.150481	0.143122

```
n_init=10: 0.2861
n_init=20: 0.2861
n_init=50: 0.2861
n_init=100: 0.2861
```

From these validation , It increases the confidence in using 4 clusters segmentation which is robust to initialization choices and superior to alternative reduced feature selection.

CLUSTER PROFILING

The standardized cluster profiles highlight the variables which differentiate the customer segments one from another . Positive values indicate above-average behaviour relative to overall customer base where negative values indicate below-average behaviour. The visualisation is much easier for interpretation of each segment by their demographic characteristics, buying habits and expense patterns.



SEGMENT INTERPRETATION

CLUSTER 0 : Occasional Traditional Customers

This segment is clustered mainly by those who don't have a loyalty card . So it can be concluded that these customers buy less and interact less with the retailer .

CLUSTER 1 : Loyal High-Value Customers

In this segment customers have loyalty cards. This suggests that these customers demonstrate high expenditure and strong engagement with the seller. They suggest long term profitability.

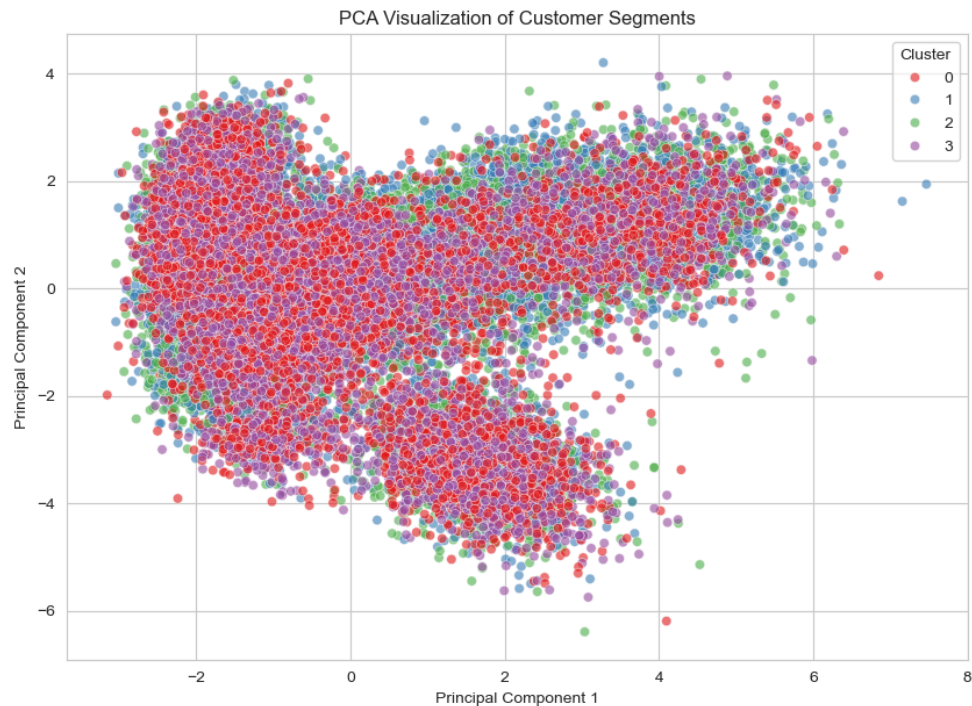
CLUSTER 2 : Family-Oriented Customers

This segment has shoppers with loyalty cards . They display high spending among household and family-related categories including hygiene products,meat and fish . The purchasing patterns indicate stability .

CLUSTER 3 : Specialised Customers

Customers in this segment shows more niche purchasing behaviour compared to the other groups. Although they dont hold loyalty cards as frequently, they demonstrate relatively higher spending on categories such as electronics and videogames

PCA



The first two principal components explained 35.1% of the total variance in the dataset, with PC1 accounting for 20.6% and PC2 accounting for 14.5% of the variance

PCA was used only as a visualization tool to explore the cluster structure in two dimensions. The result shows overlap among the four segmentation. It shows that customer behaviour is complex and cannot be fully represented using only two principal components.

```
Top variables for PC1:
lifetime_spend_fish           0.407540
lifetime_spend_meat           0.392216
lifetime_spend_alcohol_drinks 0.377087
lifetime_total_distinct_products 0.342589
lifetime_spend_nonalcohol_drinks 0.316286
lifetime_spend_electronics     0.298721
lifetime_spend_groceries       0.250531
family_size                   0.248445
lifetime_spend_videogames      0.209954
lifetime_spend_vegetables      0.186427
Name: PC1, dtype: float64

Top variables for PC2:
lifetime_spend_hygiene         0.424650
lifetime_spend_videogames      0.369382
lifetime_spend_vegetables      0.314540
typical_hour                   0.288236
lifetime_spend_electronics     0.276483
family_size                   0.259582
lifetime_total_distinct_products 0.239944
lifetime_spend_groceries       0.223333
tenure                         0.222497
lifetime_spend_nonalcohol_drinks 0.200380
Name: PC2, dtype: float64
```

The first principal component primarily captured overall purchasing intensity and product diversity, with high contributions from spending on fish, meat, alcoholic beverages, groceries, electronics, and the total number of distinct products

purchased whereas the second principal component reflected differences in lifestyle and household purchasing patterns, being strongly influenced by hygiene products, videogames, vegetables, shopping habits, electronics expenditure, and family size.

Although, PCA explained only 35% of the total variance and did not reveal the perfect separated groups the final clustering remains valid because Elbow method, silhouette analysis, reduced feature comparison, different n_iter for initializations constantly supported 4 clusters.

4/TARGETED PROMOTION

Association Rule Mining using the Apriori algorithm is employed to identify products that are frequently purchased together. The extracted rules are translated into practical marketing campaigns aimed at increasing basket size and improving customer engagement.

The strongest associations identified are presented below.

Promotion Campaign	Support	Confidence	Lift
Offer AirPods discounts to customers purchasing Bluetooth Headphones.	2.56%	34.55%	2.86
Recommend Energy Drinks to customers buying AirPods.	2.45%	33.42%	2.77
Create breakfast bundles combining Fresh Bread and Cereals.	2.26%	22.72%	2.25
Offer discounts on Eggs when Cereals are purchased.	2.20%	23.59%	2.34
Promote Tomatoes together with Asparagus through healthy meal campaigns.	2.00%	26.71%	2.09

The identified rules suggest that customers exhibit both grocery-related and electronics-related purchasing patterns. Electronics accessories such as Bluetooth Headphones and AirPods show particularly strong associations, indicating opportunities for cross-selling premium products. Similarly, breakfast items including cereals, bread and eggs frequently appeared together, supporting the development of bundle promotions..

Segment-Specific Recommendations

Cluster 0 – Occasional Traditional Customers

- **Breakfast Saver:** Buy Cereals and get **15% off Eggs**.
- **Bakery Combo:** Buy Fresh Bread and receive **10% off Butter**.
- **Come Back Bonus:** Spend over €25 and receive a **€5 voucher for your next visit**

Cluster 1 – Loyal High-Value Customers

- **Tech Upgrade Offer:** Buy Bluetooth Headphones and get **20% off AirPods**.
- **Energy Boost Deal:** Purchase AirPods and receive **25% off an Energy Drink**.
- **VIP Reward:** Loyalty card members receive **double loyalty points for one week**.

Cluster 2 – Family-Oriented Customers.

- **Family Breakfast Bundle:** Buy Cereals and Fresh Bread together and **save 15%**.
- **Family Basket Discount:** Spend over €50 and receive **10% off your entire basket**.
- **Weekend Family Deal:** Buy selected household products and receive a **free family-size product**.

Cluster 3 – Specialized Customer

- **Healthy Choice Promotion:** Buy Tomatoes and get **20% off Asparagus**.
- **Gaming Special:** Buy 1 Video Game, get the second at **50% off**.

5/ CONCLUSION

The project identified 4 meaningful customer segmentations using k-means clustering and validation through PCA, silhouette and elbow method. Association rule mining provided some relationships that were translated into promotional campaigning. The findings demonstrate how machine learning can support data driven marketing decisions and improve customer engagement strategies.

